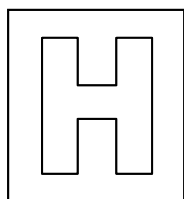


Candidate Name: _____

Class Adm No

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2013 Preliminary Examination 2 Pre-University 3

H2 Biology

9648/02

17 September 2013

2 hours

Additional Materials: Writing paper

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your Index number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer any **one** question.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question. At the end of the examination, fasten all your work securely together.

For Examiner's Use	
Section A	
1	
2	
3	
4	
5	
6	
Section B	
Total	

This question paper consists of 24 printed pages.

[Turn over

Section A

Answer **all** questions in this section.

1. A long-term breeding experiment to investigate the genetic basis of tame (friendly) behaviour was carried out in a population of silver foxes. The foxes were bred each year and the resulting young foxes assessed each month between the ages of 1 and 8 months to see how tame they were.

Table 1.1 shows how the foxes were put into categories according to their tameness.

Table 1.1

tameness class	description of behaviour towards humans
3	Not tame - these foxes run away from humans or bite when handled.
2	Neutral - these foxes allow handling by humans but show no emotionally friendly response.
1	Tame - these foxes are friendly to humans. They wag their tails and whine for attention.
elite	Very tame - these foxes are eager for human elite contact. They whimper to attract attention and sniff and lick humans.

The tamest 5% of the male foxes and the tamest 20% of the female foxes in each generation were used for breeding to produce the next generation. This was repeated for over forty generations.

(a)

- (i) State the name given to the process in which only a certain percentage of adult foxes were chosen by humans to breed in each generation.

..... [1]

- (ii) Suggest why 20% of the female foxes were used for breeding but only 5% of the male foxes.

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 [2]

- (b) Table 1.2 shows the number of foxes in the elite tameness class during the long-term experiment.

Table. 1.2

Number of generations	Foxes in elite class (%)
10	18
20	35
35	75

Discuss what the results shown in Table 1.2 suggest about the causes of variation in tameness behaviour in silver foxes.

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- (c) As tameness increased in the silver fox population over the years, it was noticed that other phenotypic traits also became more common.

Table 1.3 compares the frequency of these traits in a control group of silver foxes that had not been used in this long-term breeding experiment and in the tame population of foxes.

Table. 1.3

Phenotypic trait	Animals showing trait (per 100 000)		Percentage increase in trait
	Control Population	Tame Population	
White patch of fur on head	710	12 400	1 646
Floppy ears	170	230	35
Short tail	2	140	6 900
Curly tail	830	9 400	1 033

Students were asked to suggest a variety of genetic hypotheses to explain why these traits become more common in tame foxes. Their suggestions were:

Linkage Epistasis Inbreeding Genetic Drift

Select one hypothesis from the list and explain how it could account for the data in Table 1.3.

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- (d) Similar changes in tameness, colour and body shape are believed to have occurred in the 11 000 year period during which the grey wolf species, *Canis lupus*, evolved into the domesticated dog species, *Canis familiaris*.

Suggest how different types of isolating mechanism allowed dogs to evolve separately to wolves.

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- (e) Interbreeding between members of the wolf species and some dogs has been reported.

However, there are some large breeds of dogs that cannot breed successfully with small dog breeds.

Use this information and your own knowledge to explain the problems of classifying wolves and different dog breeds according to:

- the biological species concept

and

- the phylogenetic species concept.

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[Total: 15]

2.

- (a) The brain depends on a constant supply of oxygen for aerobic respiration. Anaerobic respiration is not sufficient to keep neurones in the brain alive. This is because neurones require especially large amounts of ATP. Up to 80% of the ATP is used to provide energy for the Na^+ / K^+ pump.

When a person suffers a stroke, blood flow to part of the brain is stopped, so some neurones receive no oxygen. ATP production by oxidative phosphorylation stops. Fig. 2.1 shows some of the ways in which the lack of ATP affects a neurone in the brain.

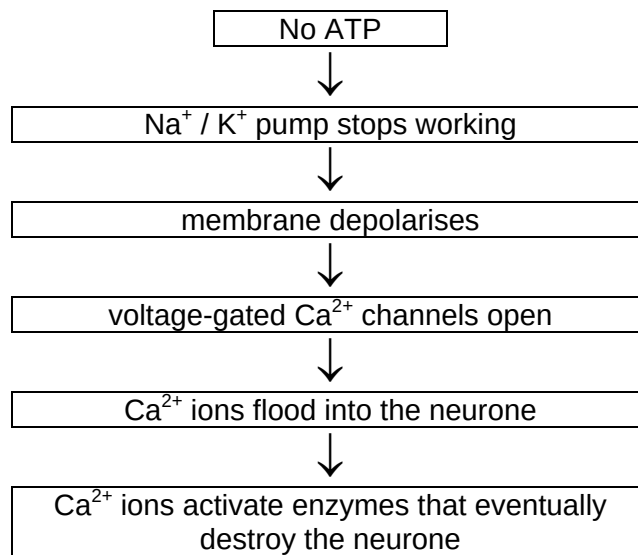


Fig. 2.1

- (i) Explain why the membrane of the neurone depolarises when the Na^+ / K^+ pump stops working.

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- (ii) Suggest why calcium ions flood into the neurone when the Na^+ / K^+ pump stops working.

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- (b) The freshwater turtle, *Trachemys scripta*, is able to survive for long periods in conditions of very low oxygen concentration. As in humans, the rate of activity of the Na^+ / K^+ pump in the neurones in its brain falls sharply. However, in turtles this does not result in damage to these cells.

A better understanding of how the neurones in the turtle's brain survive in these conditions could lead to new treatments for people who have suffered a stroke.

Experiments show that, in turtle brain neurones, in conditions of low oxygen availability:

- most ion channels in the cell surface membranes immediately close
- after about four hours, the quantity of mRNA involved in the synthesis of proteins used to build ion channels, falls to less than one fifth of normal concentrations.

- (i) Suggest how the closure of ion channels in the neurones of the turtle in very low oxygen concentrations could allow the cells to survive.

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- (ii) Suggest what causes the quantity of mRNA for protein channels to fall.

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- (c) Some synapses in the brain use the neurotransmitter dopamine. After the postsynaptic membrane has been depolarised, dopamine leaves the receptor proteins and moves back into the presynaptic neurone through specific transporter proteins.

Schizophrenia is a condition in which there is a higher than usual concentration of dopamine in certain areas of the brain. The drug phenothiazine has a similar shape to dopamine and is used to treat schizophrenia.

Suggest and explain what occurs at the synapse when phenothiazine is used in the treatment of schizophrenia.

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- (d) DRD4 is a dopamine receptor in humans. The gene coding for the DRD4 receptor has a large number of alleles, of which an individual can have only two.

Three alleles of the DRD4 receptor gene have the following mutations:

- a single base substitution
- a 21 base-pair deletion
- a 13 base-pair deletion.

The 13 base-pair deletion has the most serious consequences for the structure of the DRD4 receptor protein. Suggest why this is so.

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- (e) One allele of the *DRD4* gene has been found more frequently amongst individuals whose personality is described as 'impulsive and exploratory'.

Describe the mechanism whereby an allele such as this could have become common in the human population.

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[Total: 18]

3. Bone marrow contains stem cells that divide by mitosis to form blood cells. Each time a stem cell divides it forms a replacement stem cell and a cell that develops into a blood cell.

Fig. 3.1 shows changes in the mass of DNA in a human stem cell from the bone marrow during three cell cycles.

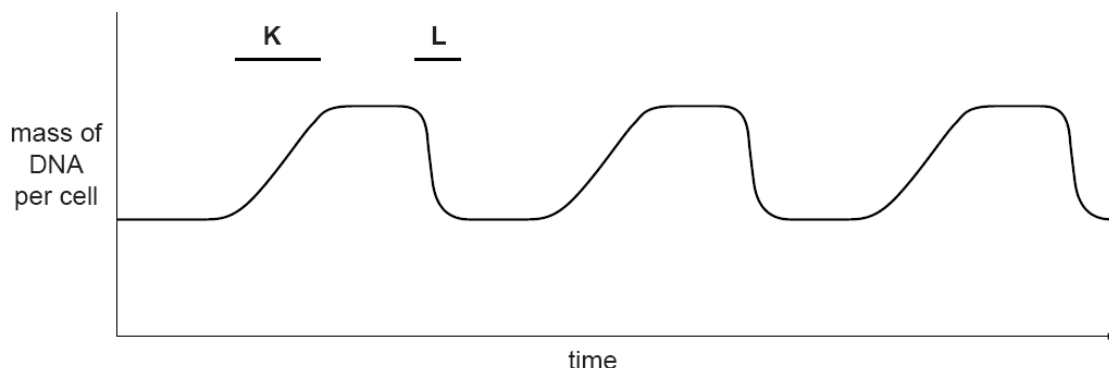


Fig. 3.1

- (a) With reference to Fig. 3.1, state:
- (i) what happens to bring about the changes in the mass of DNA per cell at **K** and at **L**.
K

L
 [2]
 - (ii) how many blood cells are formed from the stem cell in the time shown.
 [1]
 - (iii) what happens to the number of chromosomes in the stem cell.
 [1]

Stem cells in bone marrow give rise to phagocytes, B-lymphocytes and T-lymphocytes.

(b) During an immune response, cells divide by mitosis.

Suggest how mitosis is involved in an immune response.

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A protein, mitosis-promoting factor (MPF), has been identified in cells. MPF is a globular protein made from two polypeptide chains.

The presence of MPF is known to cause prophase to start.

(c) Describe the changes that occur during prophase in an animal cell.

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- (d)** MPF normally begins to break down and stops functioning during anaphase.
Suggest the possible consequences of MPF not breaking down.

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[Total: 14]

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4. A type of pheasant occurs in a range of colours, especially when bred in captivity. It may, for example, have green or purple plumage.

Sometimes when a green male is crossed with a green female all the offspring, male and female, are green. However, sometimes a green male crossed with a green female results in offspring in which the majority of the offspring are green, but in which some of the females are purple, as shown in Table 4.1.

Table. 4.1

Phenotype	Number of offspring
green male	7
green female	3
purple female	4

(a)

- (i) Using the results shown in Table 4.1, a geneticist proposed a hypothesis which predicted a 3:1 ratio of green:purple phenotypes. She used the χ^2 test to test her hypothesis

Complete Table 4.2 to calculate χ^2 .

Table. 4.2

	Green Plumage	Purple Plumage
Observed (<i>O</i>)	7 + 3 = 10	4
Expected (<i>E</i>)	10.5	3.5
$O - E$		
$(O - E)^2$		
$\frac{(O - E)^2}{E}$		
$\Sigma \frac{(O - E)^2}{E} = \chi^2$		

[2]

- (ii) Draw a conclusion from your calculation in Table 4.2, given that the critical value for χ^2 at $p = 0.05$ with one degree of freedom is 3.84. Explain why this conclusion should be treated with caution.

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..... [3]

- (b) Plumage colour in pheasants is sex-linked.

In birds, the sex chromosomes are referred to as W and Z, rather than Y and X as in mammals. The W chromosome has no genes that affect plumage colour. The heterogametic sex is the female, not the male. Thus the male has two Z chromosomes (ZZ) and the female has one W and one Z chromosome (WZ).

- (i) Use a genetic diagram to explain the results in Table 4.1.

[3]

- (ii) Using the same symbols as in (b)(i), indicate the genotypes and phenotypes of the parents which could give rise to purple male offspring.

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..... [2]

[Total: 10]

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5.

- (a) An experiment was carried out to investigate the effect of temperature on the rate of oxygen consumption of the lizard, *Sauromalus hispidus*. The body temperature of a lizard varies with environmental temperature.

Several lizards were fitted with small, airtight masks that covered their heads. Air was supplied inside the mask through one tube, and collected through another. The differences between oxygen concentrations in the air supplied for inhalation and the exhaled air enabled the researchers to measure the rate of oxygen consumption of the lizards.

The rate of oxygen consumption of each lizard was measured when it was at rest and when it was running. Measurements were made at different temperatures ranging from 15 °C to 40 °C.

Fig. 5.1 shows the results.

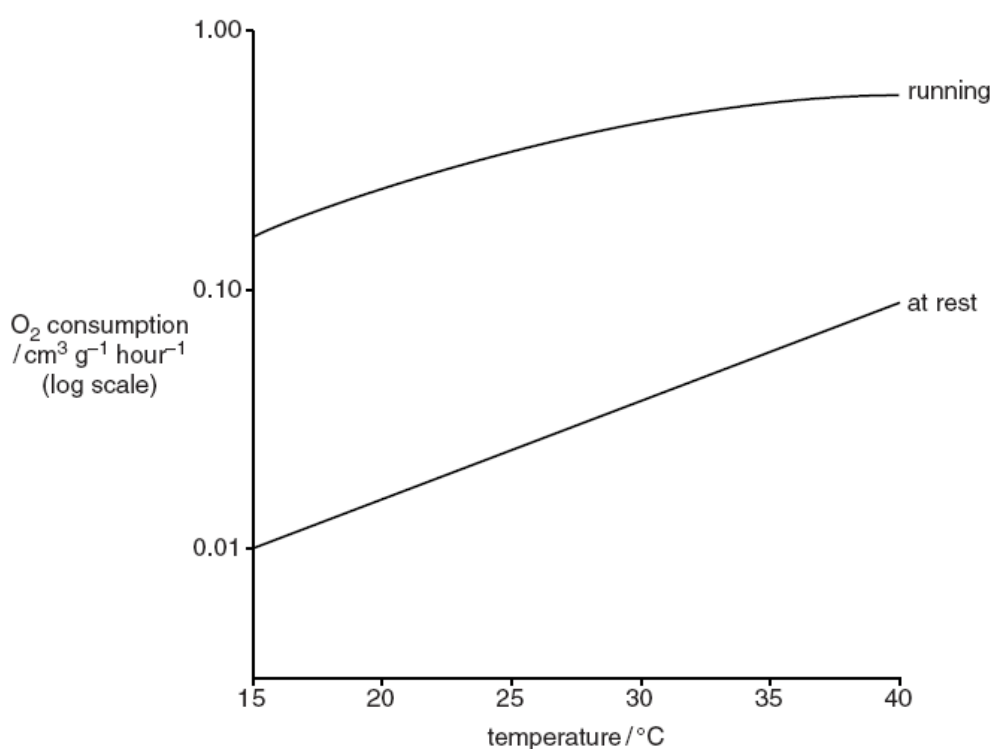


Fig. 5.1

- (i) Running requires rapid use of ATP by muscle cells in the legs and heart of a lizard.

With reference to the events occurring inside a mitochondrion, explain why a faster use of ATP requires a greater rate of oxygen consumption.

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- (ii) Explain the effect of temperature on the rate of oxygen consumption in *Sauromalus* when at rest.

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- (b) In the light-independent stage of photosynthesis, the enzyme rubisco catalyses the combination of carbon dioxide with ribulose biphosphate, RuBP. When the carbon dioxide concentration within the leaf is very low, rubisco tends to combine oxygen, rather than carbon dioxide, with RuBP. This process is called photorespiration. It reduces carbon dioxide assimilation and therefore reduces crop yields.

Photorespiration is most likely to happen in hot, dry conditions.

Suggest why photorespiration is most likely to take place in hot, dry conditions

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- (c) It is expected that the carbon dioxide concentration in the atmosphere will increase in the future, which would be expected to increase rates of photosynthesis in many crop plants.

Investigations were carried out into the effect of increased carbon dioxide concentration on the rate of photosynthesis in maize.

- Maize plants were grown in open-air trials, in the same field and were exposed to the same changes in the weather.
- 50% of the plants were exposed to a normal carbon dioxide concentration.
- 50% of the plants were exposed to an increased carbon dioxide concentration.
- The rate of photosynthesis was measured as the net assimilation rate of carbon dioxide.
- Measurements were made at three-hourly intervals between 0700 hours and 1900 hours on three different days.

The results are shown in Fig. 5.2.

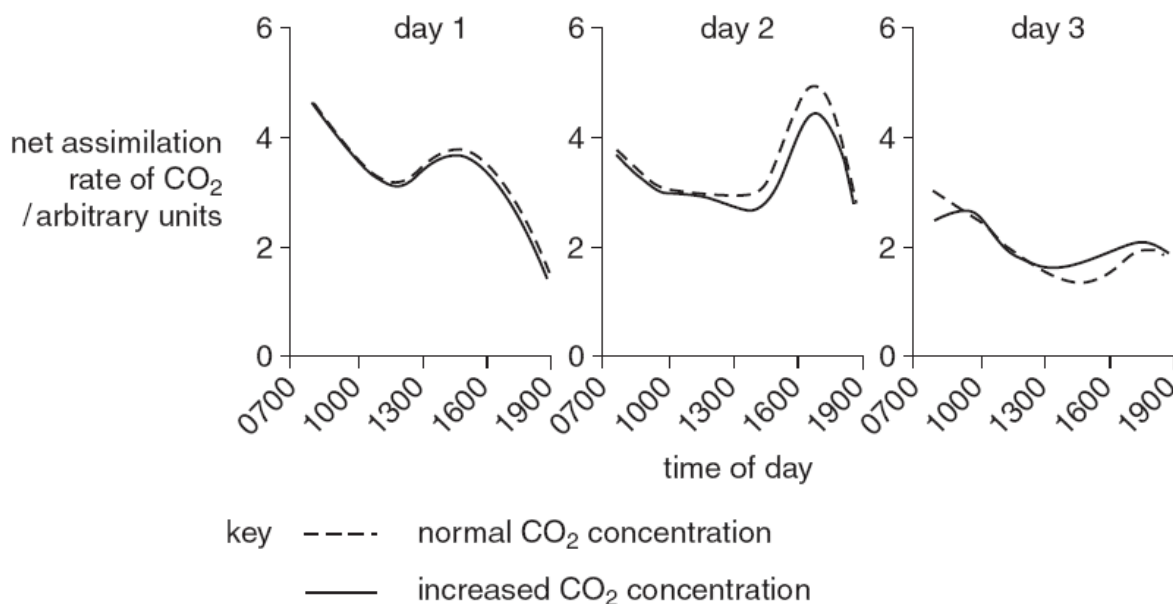


Fig. 5.2

- (i) Suggest an explanation for the lack of effect of carbon dioxide concentration on the rate of photosynthesis in maize plants, shown by these results.

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 [2]

- (ii) Suggest one explanation for the changes in the rate of photosynthesis between 0700 hours and 1900 hours on day 1.

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 [2]

[Total: 14]

6. A study finding published in the journal PLoS Genetics in 2013 was the first to show exercise alters DNA methylation related to fat storage.

The finding adds to a growing body of knowledge about epigenetics that show though we cannot change our DNA we can change the way our genes are expressed to fight disease.

- (a) Describe and explain the effect of DNA methylation in eukaryotic genome.

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- (b) Suggest how DNA methylation enables cells to differentiate.

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..... [1]

In the 1950s, Erwin Chargaff determined the relative quantities of the four bases in DNA in different organisms. His results provided important evidence for the model of DNA proposed by James Watson and Francis Crick in 1953. Some of Chargaff's data is shown in Table 6.1.

Table. 6.1

organism	percentage of adenine	percentage of thymine	percentage of guanine	percentage of cytosine
<i>Escherichia coli</i> (bacterium)	24.7	23.6	26.0	25.7
a yeast	31.3	32.9	18.7	17.1
wheat	27.3	27.1	22.7	22.8
octopus	33.2	31.6	17.6	17.6
sea urchin	32.8	32.1	17.7	17.3
chicken	28.0	28.4	22.0	21.6
human	29.3	30.0	20.7	20.0

- (c) Explain how the data in Table 6.1 helps to confirm the arrangement of bases in DNA.

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- (d) Table 6.2 shows Chargaff's data for a virus.

Table. 6.2

organism	percentage of adenine	percentage of thymine	percentage of guanine	percentage of cytosine
a virus	24.0	31.2	23.3	21.5

- (i) State how the result for the virus differs from the results for all the organisms given in Table 6.1.

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- (ii) Suggest why the results for the virus are different from all the other organisms.

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[Total: 9]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

7.

(a) Outline the molecular structure of phospholipids in relation to their function in cell membranes. [7]

(b) Explain the role of the Golgi body and its link to the rough endoplasmic reticulum. [7]

(c) Describe the structure of ribosomes. [6]

[Total: 20]

8.

(a) Explain the effects of competitive and non-competitive inhibitors on the rate of enzymatic activity. [6]

(b) Explain the mode of action of enzymes in terms of specificity and activation energy. [10]

(c) Describe the effect of temperature on enzyme-catalyzed reactions. [4]

[Total: 20]